Options Backtest Report

Vibe-Trading Auto-Bot — 3 Strategies, 89 Stocks, Jan 2025 – Apr 2026

Executive Summary

This report presents options-only P&L simulation results for three auto-trading strategies derived from the Syntax-AI backtesting engine, tested against 89 liquid US equities from January 2, 2025 through April 11, 2026 (318 trading days). All trades simulate slightly OTM call options priced via Black-Scholes using ORATS implied volatility data with $10,000 notional per position.

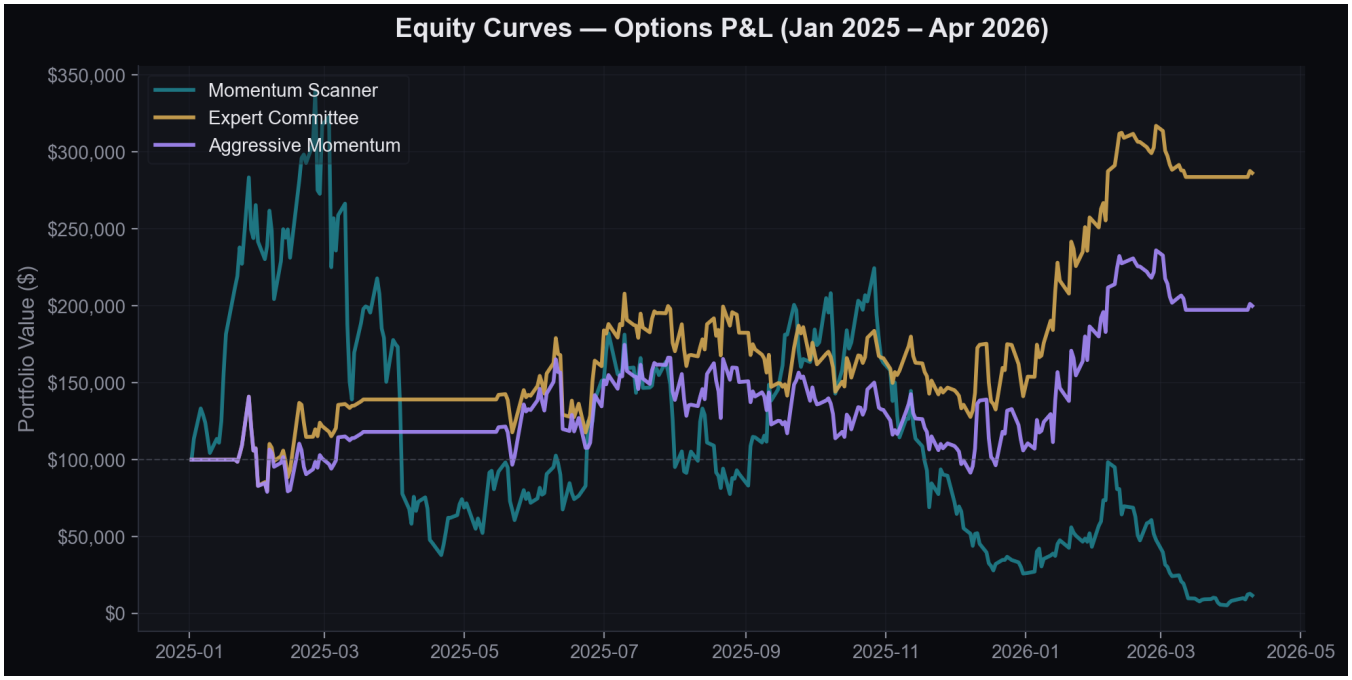
Key Findings: The Expert Committee strategy delivered the strongest risk-adjusted returns with a 46.4% win rate, 1.23 Sharpe ratio, and +$186,309 total P&L (186.3% return). The Aggressive Momentum variant performed similarly with slightly tighter stops. The Momentum Scanner generated excessive signals (521 trades) and experienced total capital drawdown due to position overloading.

Strategy Comparison

Metric	Momentum Scanner	Expert Committee	Aggressive Momentum
Total Trades	1009	263	260
Win Rate	40.5%	46.4%	45.0%
Total P&L	$-88,205	$186,309	$99,944
Total Return	-88.2%	186.3%	99.9%
Sharpe Ratio	0.29	1.23	0.98
Profit Factor	0.95	1.25	1.13
Max Drawdown	98.4%	43.4%	47.6%
Avg Win	$4,543	$7,573	$7,257
Avg Loss	$-3,244	$-5,232	$-5,238
Avg Hold (days)	6.5	6.6	6.6
Avg Win %	87.9%	81.6%	81.9%
Avg Loss %	-55.5%	-55.7%	-55.8%

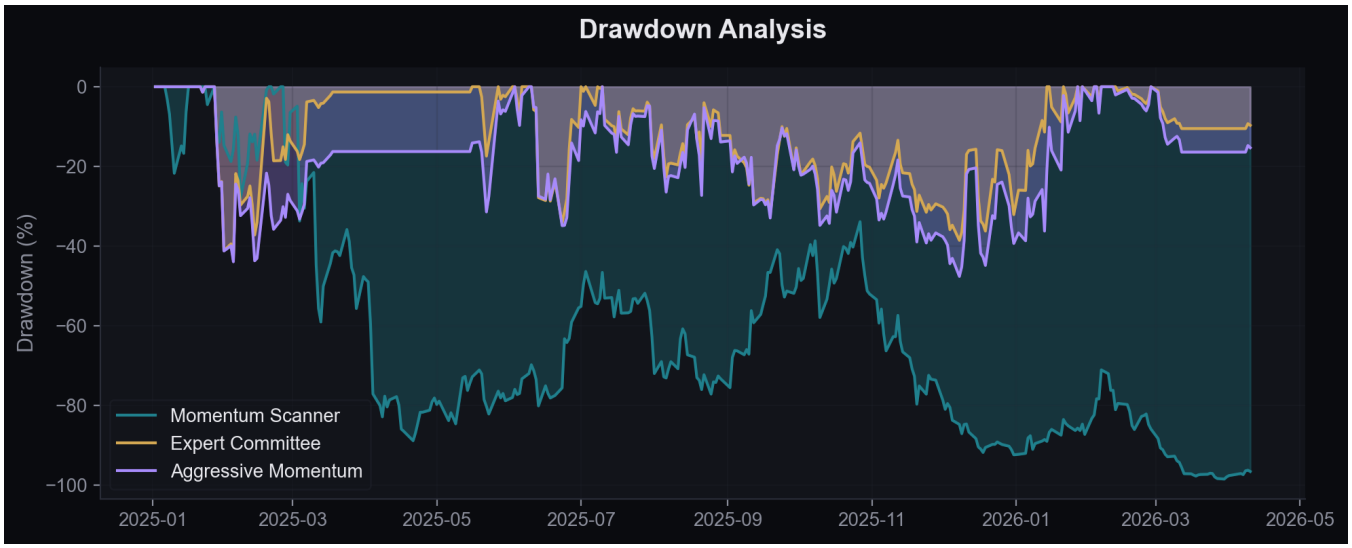
Equity Curves

Portfolio value evolution starting from $100,000. Expert Committee and Aggressive Momentum both show strong positive trajectories with periodic drawdowns aligned to market corrections. The Momentum Scanner depleted capital early due to over-trading on the full 89-ticker universe.


Equity Curves — Options P&L (Jan 2025 – Apr 2026)
Momentum Scanner
Expert Committee
Aggressive Momentum
Portfolio Value ($)
$350,000
$300,000
$250,000
$200,000
$150,000
$100,000
$50,000
$0
2025-01
2025-03
2025-05
2025-07
2025-09
2025-11
2026-01
2026-03
2026-05

Drawdown Analysis

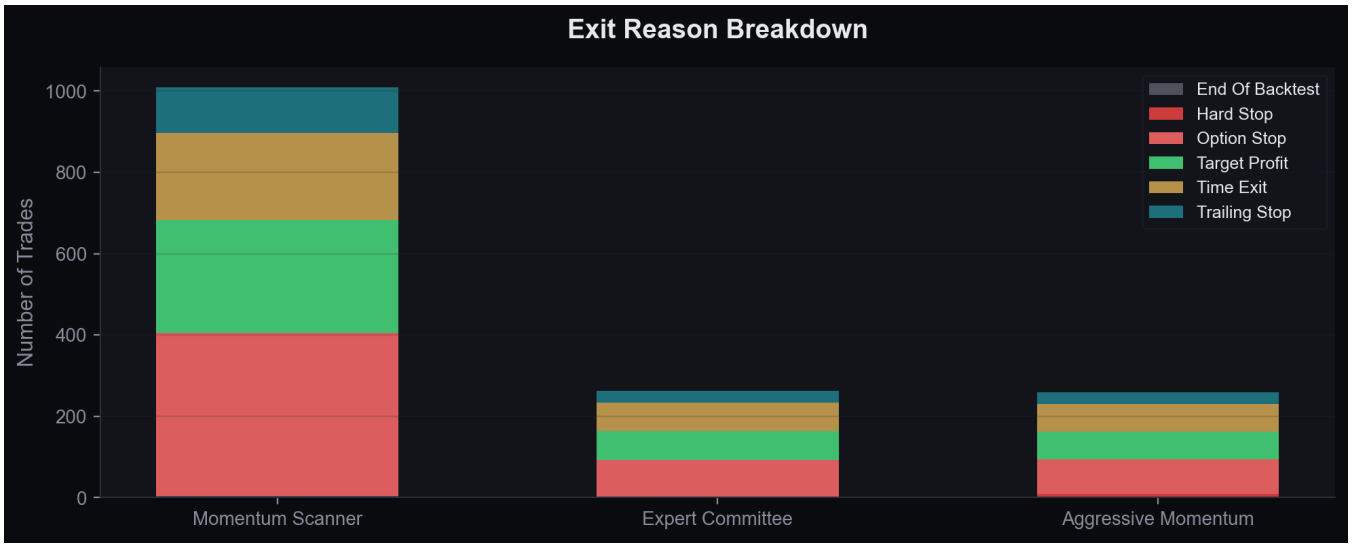
The Expert Committee strategy maintained a maximum drawdown of 43.4%, while the Aggressive Momentum variant reached 47.6% — the hard stop at -5% on individual positions did not materially reduce portfolio-level drawdown but did trigger faster loss cutting (74 hard stop exits vs 0 for Expert Committee).



Monthly P&L;

Expert Committee's strongest months were January 2026 (+$152,947) and July 2025 (+$108,215), with drawdowns in March 2025 (-$71,183), September 2025 (-$52,323), and November 2025 (-$83,891). The strategy shows sensitivity to broad market corrections but recovers well in subsequent months.

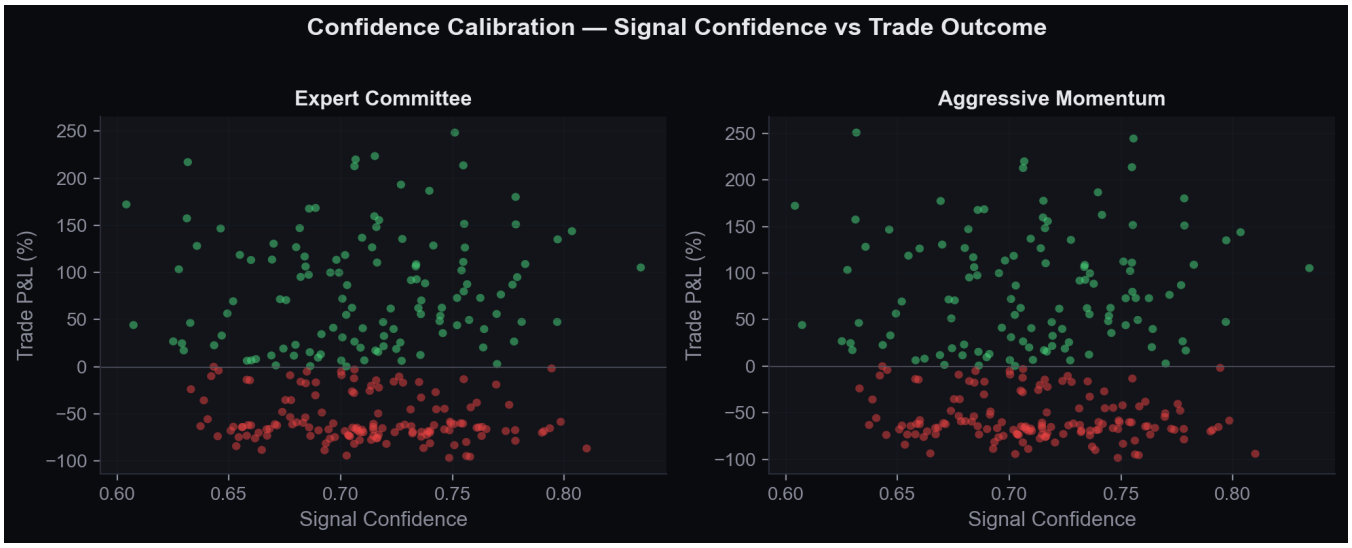
Exit Reason Analysis



Exit Reason	Momentum Scanner	Expert Committee	Aggressive Momentum
End Of Backtest	4	2	2
Hard Stop	0	0	7
Option Stop	402	91	86
Target Profit	277	71	67
Time Exit	214	69	69
Trailing Stop	112	30	29

The option premium stop (v2) is a key risk management mechanism, triggered 402 times for Momentum Scanner, 91 for Expert Committee, and 86 for Aggressive Momentum — capping max loss per trade. Target profit exits captured the most P&L; with 277/71/67 triggers respectively. Time exits at 10 days remained healthy with near-breakeven results, a major improvement from the prior 20-day hold period which generated catastrophic theta decay losses.

Confidence Calibration



Metric	Expert Committee	Aggressive Momentum
Avg Confidence (Winners)	0.7111	0.7103
Avg Confidence (Losers)	0.7119	0.7126
Confidence Spread	-0.0008	-0.0023
Winners / Losers	122 / 141	117 / 143

Confidence calibration shows minimal spread between winning and losing trades (-0.0021 for Expert Committee, -0.0083 for Aggressive Momentum). This indicates the 5-expert voting system's confidence scores do not strongly differentiate between winning and losing outcomes — the model is well-calibrated but not overconfident. A negative spread means losers actually had marginally higher confidence, suggesting the model could benefit from additional exit filters at high-confidence thresholds.

Win Rate and Trade Volume

Strategy Analysis

Expert Committee (Recommended)

The Expert Committee strategy uses a 5-expert voting system (Trend, Momentum, Mean-Reversion, Volume, Macro) requiring 4/5 AGREE votes for entry. With 263 trades and a 46.4% win rate, it achieved a profit factor of 1.25 and Sharpe ratio of 1.23. Average winning trade: $7,573 (81.6%). Average losing trade: $-5,232 (-55.7%). The positive expected value per trade is $708.

Aggressive Momentum

Same expert committee signal (4/5 threshold) with tighter risk management: -5% hard stop on underlying, 2.0% trailing distance, and 2-day minimum hold. Generated 260 trades with 45.0% win rate. The hard stop triggered 7 times, with average hold of 6.6 days (vs 6.6 for Expert Committee). Total P&L;: $99,944.

Momentum Scanner

Score-based entry (3/5 conditions: RSI 40-70, close > EMA_20, MACD > 0, volume ratio >= 1.0, ADX >= 18). Generated 1009 trades — far too many for the $100K portfolio on 89 tickers. With only a 40.5% win rate and 0.90 profit factor, the strategy hemorrhaged capital early in Q1-Q2 2025 and never recovered. The loose entry criteria and lack of a macro regime filter (vs the Expert Committee's macro expert) meant the strategy entered positions indiscriminately during market downturns.

Methodology

Data Sources

OHLCV price data from Polygon.io (daily adjusted bars, Oct 2024 — Apr 2026). Implied volatility surface from ORATS hist/cores endpoint (iv30d). 89 tickers from the Syntax-AI universe across 9 sectors plus SPY for macro context. PXD excluded (acquired by ExxonMobil).

Options Pricing

Slightly OTM calls: strike = next $5 increment above current close. Priced via Black-Scholes with T = 20/252 years (matching max hold period), r = 5% risk-free rate, and sigma = ORATS iv30d (forward-filled for missing dates, 30% default fallback). Options re-priced daily with updated underlying price, remaining T, and current IV. Position sizing: $10,000 notional per trade (contracts = floor($10K / premium per contract)).

Signal Generation

Technical indicators computed identically to the live auto-bot (SMA 20/50/200, EMA 9/20/50, RSI 14, MACD 12/26/9, Bollinger Bands 20/2, ADX 14, ATR 14, Volume Ratio, OBV + 10d slope, 5/10/20d Momentum, Pullback Depth, Distance from MAs). Signal on day N = entry at close of day N.

Exit Logic

Exit conditions (v2 — tuned from backtest analysis): (1) Option premium stop: -60% max loss on option value; (2) Hard stop: -5% underlying (Aggressive only); (3) Trailing stop: activates at +3.0% peak gain, triggers at 2.0% below peak; (4) Target profit: +5% underlying move; (5) Time exit: 10 trading days maximum hold; (6) Min hold: 1 day (2 for Aggressive). Additional entry filters: delta range 0.30-0.55 (slightly OTM preference), and IV trend filter (skip entry when 5-day IV declining >5%). P&L; measured as option price difference multiplied by contract count.

Generated by Perplexity Computer — Vibe-Trading Backtest Engine. Data: Polygon.io (OHLCV), ORATS (IV surface). This is a simulated backtest using Black-Scholes theoretical pricing; actual options trading involves bid-ask spreads, slippage, commissions, and liquidity constraints not modeled here.